

Cobalt carbide nanoprisms for direct production of lower olefins from syngas

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Lower olefins—generally referring to ethylene, propylene and butylene—are basic carbon-based building blocks that are widely used in the chemical industry, and are traditionally produced through thermal or catalytic cracking of a range of hydrocarbon feedstocks, such as naphtha, gas oil, condensates and light alkanes^{1,2}. With the rapid depletion of the limited petroleum reserves that serve as the source of these hydrocarbons, there is an urgent need for processes that can produce lower olefins from alternative feedstocks^{3–9}. The ‘Fischer–Tropsch to olefins’ (FTO) process has long offered a way of producing lower olefins directly from syngas—a mixture of hydrogen and carbon monoxide that is readily derived from coal, biomass and natural gas^{3–7}. But the hydrocarbons obtained with the FTO process typically follow the so-called Anderson–Schulz–Flory distribution, which is characterized by a maximum C₂–C₄ hydrocarbon fraction of about 56.7 per cent and an undesired methane fraction of about 29.2 per cent (refs 1, 10–12). Here we show that, under mild reaction conditions, cobalt carbide quadrangular nanoprisms catalyse the FTO conversion of syngas with high selectivity for the production of lower olefins (constituting around 60.8 per cent of the carbon products), while generating little methane (about 5.0 per cent), with the ratio of desired unsaturated hydrocarbons to less valuable saturated hydrocarbons amongst the C₂–C₄ products being as high as 30. Detailed catalyst characterization during the initial reaction stage and theoretical calculations indicate that preferentially exposed {101} and {020} facets play a pivotal role during syngas conversion, in that they favour olefin production and inhibit methane formation, and thereby render cobalt carbide nanoprisms a promising new catalyst system for directly converting syngas into lower olefins.

The catalytic mechanism for the FTO process is very similar to that for typical Fischer–Tropsch synthesis, and Fischer–Tropsch catalysts can be used to produce lower olefins after proper catalyst modification and optimization of reaction conditions^{3–7,13–17}. Recently, it was reported that iron-based Fischer–Tropsch catalysts promoted by sulfur

and sodium exhibited excellent selective formation of lower olefins (which constituted 61% of the carbon products, that is, 61 C%)³. Nonetheless, with the goal of simultaneously achieving high selectivity for the production of lower olefins, low selectivity for methane production, and high stability, it is necessary to develop new FTO catalysts that operate away from the Anderson–Schulz–Flory (ASF) distribution.

Here, we prepared cobalt–manganese composite oxide (CoMn catalyst) and investigated it in the FTO reaction (Table 1). This catalyst, after reaching the steady state, displayed a high selectivity for the production of lower olefins (60.8 C%) and a low methane selectivity (5.0 C%) at a CO conversion of 31.8% under mild reaction conditions (250 °C, 1 bar, H₂/CO ratio of 2). This type of catalyst was even suitable for converting coal- and biomass-derived syngas with low H₂/CO ratios. When the H₂/CO ratio was decreased from 2 to 0.5, methane selectivity decreased to 2.4 C%, and the olefin/paraffin ratio for C₂–C₄ slate increased to 51, while selectivity for the production of lower olefins remained high (45.1 C%). (A high olefin/paraffin ratio is of significance industrially; when the ratio is high enough, the paraffin can be ignored and the product used as if it were pure olefins.) The product distribution deviated greatly from the classical ASF distribution, with the highest selectivity being for propylene (Extended Data Fig. 1). Higher total pressures increased the selectivity for the production of oxygenates and decreased the olefin/paraffin ratio, owing to the enhanced second hydrogenation of olefins (Table 1). Obviously, a lower H₂/CO ratio (≤1) and low reaction pressure (≤5 bar) resulted in greater olefin formation. We also carried out a stability test under optimized reaction conditions from an industrial viewpoint; there was no obvious deactivation over 600 h or more (Extended Data Fig. 2), suggesting promising potential industrial application.

These steady-state results differ from the typical performance of a cobalt-based Fischer–Tropsch catalyst. Even more noteworthy is how the catalytic performance evolved during the reaction (Fig. 1a and Extended Data Table 1). Over the first 20 h, both the CO-conversion rate and the product selectivity changed considerably. At the beginning,

Table 1 | Catalytic performance of the CoMn catalyst at different H₂/CO ratios and reaction pressures

Pressure (bar)	H ₂ /CO ratio	CO conversion (C%)	CO ₂ selectivity (C%)	Product selectivity (C%, CO ₂ -free)					Olefin/paraffin ratio			
				CH ₄	C ₂ –4 [≡]	C ₂ –4 ^o	C ₅ +	Oxy.	C ₂	C ₃	C ₄	C ₂ –4
1	2	31.8	47.3	5.0	60.8	2.0	31.4	0.8	19	41	29	30
1	1	11.5	48.0	3.7	50.0	1.3	43.5	1.5	27	53	36	40
1	0.5	6.3	48.3	2.4	45.1	0.9	48.7	2.0	35	68	46	51
3	0.5	14.3	48.4	4.2	44.3	2.9	38.6	10.0	7	26	20	15
5	0.5	23.6	48.0	3.7	41.2	4.7	35.7	14.7	4	13	10	8
10	0.5	28.6	46.6	4.6	31.9	5.7	39.1	18.6	2	8	6	5

Reaction conditions: 250 °C, 2,000 ml h^{−1} g_{cat}^{−1} (where g_{cat} denotes grams of catalyst). Oxy., oxygenates.

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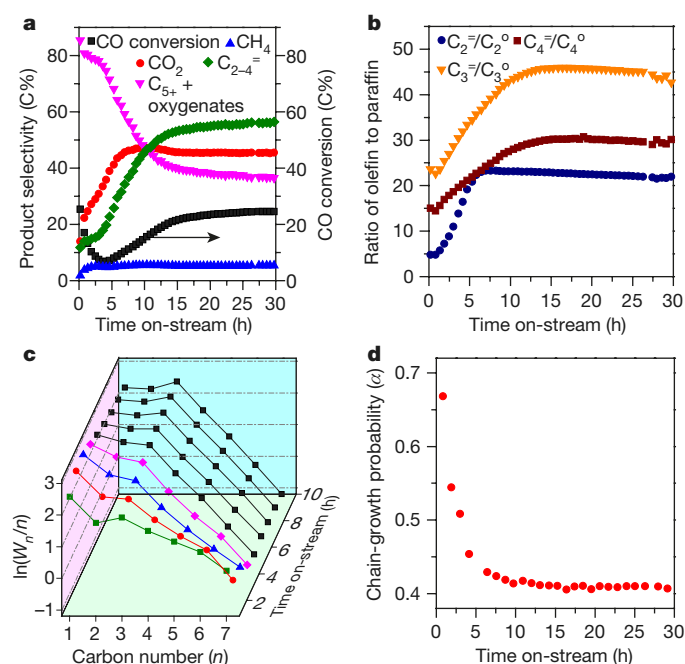


Figure 1 | Catalytic performance of the CoMn catalyst in the initial stages of the reaction. **a**, CO conversion and product selectivity as a function of time on-stream. Products are CH_4 , CO_2 , C_{2-4} , and C_{5+} + oxygenates. **b**, Ratio of olefin to paraffin as a function of time on-stream. **c**, Product plots ($\ln(W_n/n)$ and n) versus time on-stream. W_n is the fraction, by weight, of a product with n carbon atoms. **d**, Probability of the growth of carbon chains (α) as a function of time on-stream, obtained by fitting the results generated for chains of three to seven carbons using the ASF model. Reaction conditions: 250°C , 1 bar, $2,000\text{ ml h}^{-1}\text{ g}_{\text{cat}}^{-1}$, $\text{H}_2/\text{CO} = 2$.

after the syngas feed was introduced, high activity and high C_{5+} selectivity were achieved, while the selectivity for light hydrocarbons and CO_2 was rather low. Over the next 4 h, CO conversion dramatically decreased and then reached a minimum value (6.8% at 4 h). During this stage (0–4 h), the selectivity for CO_2 increased gradually. Regarding the distribution of the hydrocarbon products, methane production first increased and then remained stable at around 5 C%; meanwhile, the production of lower olefins increased continuously at the expense of C_{5+} production. In the following stage (4–15 h), the catalytic activity increased markedly and the product distribution shifted greatly from heavy hydrocarbons to lower olefins. After 15 h, the catalytic performance seemed to be stable, and no obvious change was observed afterwards. CO conversion stabilized at about 24% and CO_2 selectivity remained at about 45 C%. Amongst the hydrocarbon products, the selectivity for lower olefins and methane remained at about 55 C% and 5 C%, respectively. The olefin/paraffin ratio also increased gradually with time on-stream, and then became stable at 15 h (Fig. 1b). At steady state, the ratios of $\text{C}_2^=/\text{C}_2^0$, $\text{C}_3^=/\text{C}_3^0$ and $\text{C}_4^=/\text{C}_4^0$ were 22, 45 and 30, respectively (where $\text{C}_n^=$ is an olefin product with n carbons, and C_n^0 is a paraffin product with n carbons).

We plot the distribution of hydrocarbon products as a function of time on-stream in Fig. 1c. The $\ln(W_n/n)$ value of C_1 was higher than that of C_2 at the beginning (W_n is the fraction, by weight, of a carbon product with n carbon atoms). However, the $\ln(W_n/n)$ values of C_1 and C_2 were similar after 10 h. The $\ln(W_n/n)$ value of C_3 was the highest, consistent with the highest selectivity for C_3 product. Generally, the $\ln(W_n/n)$ value of C_1 showed a substantial deviation from the typical ASF distribution and was much lower than the modelled value. Figure 1d shows the variation in the probability of chain growth (α) for hydrocarbons as a function of time on-stream; this probability decreased rapidly within the first 4 h and came to a steady value gradually, corresponding to the variation in hydrocarbon selectivity.

As the performance of the catalyst studied here is very different from that of the typical cobalt-based Fischer–Tropsch catalyst, we reasoned that the active site might not be metallic cobalt. In order to reveal the nature of the active site that favours the formation of lower olefins, and to elucidate the structure–performance relationship, we used X-ray diffraction (XRD) to investigate the structure of spent catalyst samples at different stages of the reaction (Extended Data Fig. 3). After the syngas feed was introduced, the metallic cobalt increasingly turned into cobalt carbide (Co_2C) with time on-stream. In the meantime, cobalt and manganese, initially present as a single phase ($\text{Co}_x\text{Mn}_{1-x}\text{O}$), segregated into Co_2C and MnO , respectively. Quantitative analysis of the different phases (Co , Co_2C , $\text{Co}_x\text{Mn}_{1-x}\text{O}$ and MnO) at different stages of the reaction is shown in Extended Data Fig. 3h. For the freshly reduced sample, the relative contents of metallic Co and $\text{Co}_x\text{Mn}_{1-x}\text{O}$ were 3.9% and 96.1%, respectively. After 2 h of reaction, almost all of the metallic Co was transformed into Co_2C ; there was also a small amount of MnO . As the reaction proceeded, the amount of Co_2C and MnO increased further while that of $\text{Co}_x\text{Mn}_{1-x}\text{O}$ decreased gradually. The existence of metallic Co in the fresh reduced catalyst was responsible for the low CO_2 selectivity, low methane selectivity and high selectivity for long-chain hydrocarbons during the initial reaction stage (before 4 h). The fact that the Co_2C content increased with reaction time suggests that Co_2C might be the active phase for the FTO reaction.

It is often considered that the formation of Co_2C is responsible for the deactivation of cobalt-based Fischer–Tropsch catalysts^{18–23}, in these studies, the Co_2C nanoparticles exhibited nanosphere-like morphology. Therefore, we prepared Co_2C nanoparticles with nanosphere-like morphology as described¹⁸, and tested them for syngas conversion (Extended Data Fig. 4). We found relatively low CO-hydrogenation activity and very high selectivity to methane (about 80 C%) for such Co_2C nanospheres, consistent with ref. 18. However, the Co_2C formed from our CoMn catalyst exhibited much higher activity and promising product selectivity during the FTO reaction, suggesting that there could be some essential differences in the form of the Co_2C phase. Representative transmission electron micrograph (TEM) pictures of our as-prepared CoMn catalyst are shown in Fig. 2a–e and Extended Data Fig. 5a–d, and indicate that some nanoparticles presented a quadrangular nanoprism morphology with the shape of a parallelepiped. Further high-resolution (HR)-TEM characterizations and TEM-energy-dispersive X-ray (EDX) analyses demonstrate that such nanoprisms were composed of Co_2C , with specific exposed facets of planar geometry (101), (-101) and (020) (Fig. 2c–e and Extended Data Fig. 5e–h). In addition, sphere-like nanoparticles were composed of MnO or $\text{Co}_x\text{Mn}_{1-x}\text{O}$ (Extended Data Fig. 5c, d). The geometric model in Fig. 2f illustrates that the Co_2C nanoprisms had a parallelepiped morphology with a pair of rhomboid faces. We ascribe the pair of rhomboid faces to the facets of (020), and the other rectangles to the facets of (101) and (-101). On the basis of TEM observations at different reaction times (Extended Data Fig. 6), we suggest that the Co_2C nanocrystals were formed and gradually evolved into a nanoparallelepiped shape as the initial reaction proceeded (especially from 0 h to 10 h)—consistent with the change in catalytic performance.

Alkali elements and manganese have a ‘promoter’ effect, improving the activity and selectivity of Fischer–Tropsch synthesis^{6,24–27}. For our CoMn catalyst, prepared by co-precipitation using sodium carbonate, the residual sodium (0.33 wt%, as judged by inductively coupled plasma (ICP) mass spectrometry; Extended Data Fig. 7) and manganese might have a large effect on the catalyst structure and catalytic performance. On the basis of a control study (Extended Data Figs 7, 8), we suggest that the residual sodium enhances the formation of Co_2C , while the manganese contributes to the formation of nanoprisms, via the $\text{Co}_x\text{Mn}_{1-x}\text{O}$ precursor. As the catalytic performance of our Co_2C nanoprisms was substantially different from that of reported sodium (or manganese)-promoted, iron (or cobalt)-based Fischer–Tropsch catalysts, we suggest that Co_2C nanoprisms with exposed facets of {101} and {020} geometry represent a new active phase for syngas conversion.

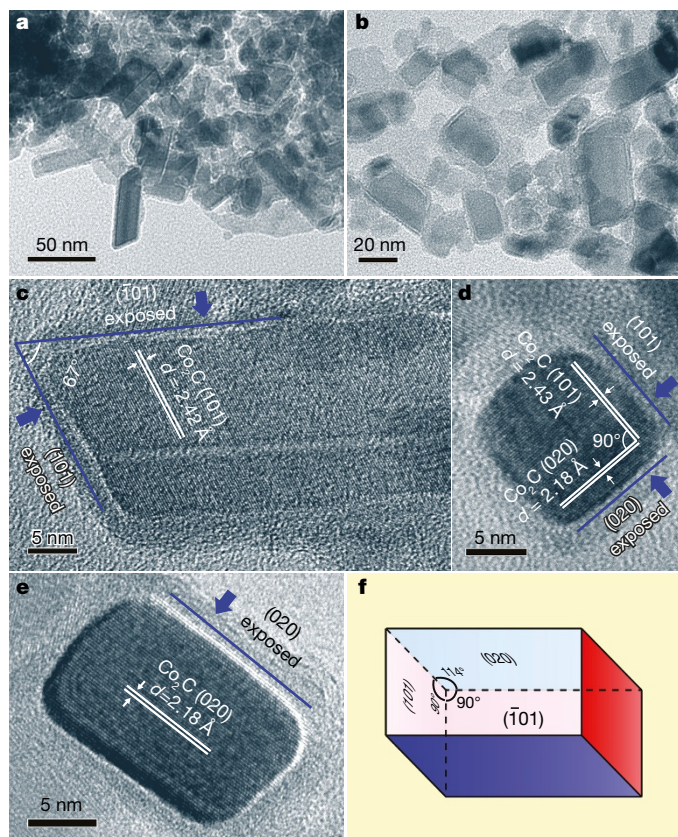


Figure 2 | TEM images of the CoMn catalysts after reaching steady state. **a, b**, Low-resolution TEM images. **c–e**, High-resolution images of Co_2C nanoprisms with exposed facets of $\{101\}$, $\{-101\}$ and $\{020\}$. **d**, distance (length) of the lattice fringes. **f**, The Co_2C nanoprism has a parallelepiped shape, with four rectangular faces and two rhomboid faces.

To shed further light on the origin of the promising catalytic performance of the Co_2C nanoprisms, we resorted to density functional theory (DFT) calculations of the formation of lower olefins (CH_2CH_2 as an example) and of methane on $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$, $\text{Co}_2\text{C}(111)$ and $\text{Co}(0001)$ surfaces (Extended Data Fig. 9a). Taking $\text{CH}_x + \text{CH}_x + \text{H}$ (where $x = 2$ or 3) as starting points, we examined three possible paths to CH_3CH_3 formation on the $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$, $\text{Co}_2\text{C}(111)$ and $\text{Co}(0001)$ surfaces, and two paths to CH_2CH_2 formation. From the energy profiles in Fig. 3, we can see that the major chain-growth pathways vary from one surface to the other and are associated with different kinetic barriers and reaction thermodynamics. For $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$, $\text{Co}_2\text{C}(111)$ and $\text{Co}(0001)$, the calculated barriers for CH_2CH_2 formation via coupling of CH_2 are 0.52 eV, 1.48 eV, 1.31 eV and 0.41 eV, respectively. The overall reaction energies are -0.52 eV, 0.57 eV, 0.16 eV and -0.47 eV, respectively. The effective barriers (ΔE_{eff}) for the formation of CH_3CH_3 from the zero point ($\text{CH}_2 + \text{CH}_2$) are 0.73 eV, 2.50 eV, 1.57 eV and 0.44 eV, respectively. Figure 3a–c also shows that CH_2CH_2 is the most stable intermediate on the $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$ and $\text{Co}_2\text{C}(111)$ surfaces during the overall pathways; CH_3CH_3 is the most stable intermediate on $\text{Co}(0001)$ (Fig. 3d). We also investigated the hydrogenation pathway from CH_x ($x = 2$ or 3) species to methane. The effective barriers for methane formation from CH_2 and H on $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$, $\text{Co}_2\text{C}(111)$ and $\text{Co}(0001)$ surfaces are 0.89 eV, 1.87 eV, 1.39 eV and 0.82 eV, respectively (Extended Data Fig. 9b). Methane formation is exothermic only on the $\text{Co}(0001)$ surface.

On the basis of these DFT calculations, we conclude that only on the $\text{Co}_2\text{C}(101)$ surface does CH_2CH_2 remain the most stable species, from both a thermodynamic and a kinetic point of view. Meanwhile, it is difficult to form methane on both $\text{Co}_2\text{C}(101)$ and $\text{Co}_2\text{C}(020)$. So, these

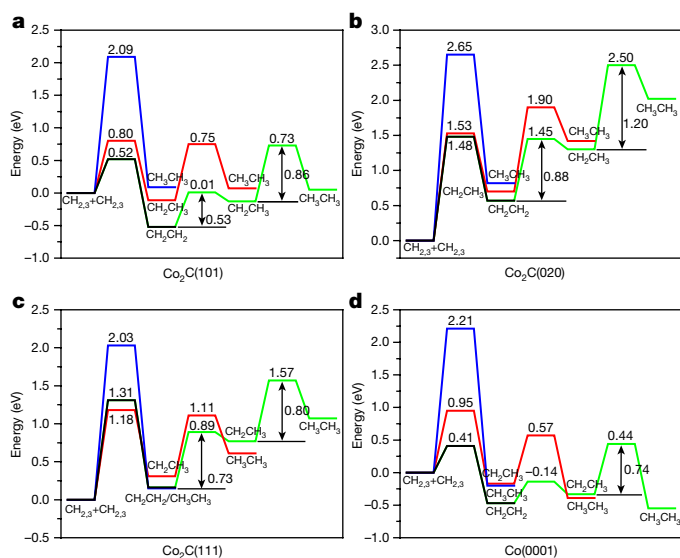


Figure 3 | Energy profiles for pathways that lead to the formation of CH_2CH_2 and CH_3CH_3 on different surfaces of Co_2C and Co . **a**, The $\text{Co}_2\text{C}(101)$ surface. **b**, The $\text{Co}_2\text{C}(020)$ surface. **c**, The $\text{Co}_2\text{C}(111)$ surface. **d**, The $\text{Co}(0001)$ surface. The intermediate state of $\text{CH}_x + \text{CH}_x$ (where $x = 2$ or 3) on the surfaces are chosen as the zero point for all the energy profiles. The path in black is $\text{CH}_2 + \text{CH}_2 \rightarrow \text{CH}_2\text{CH}_2$; the path in green is $\text{CH}_2 + \text{CH}_2 + 2\text{H} \rightarrow \text{CH}_2\text{CH}_2 + 2\text{H} \rightarrow \text{CH}_2\text{CH}_3 + \text{H} \rightarrow \text{CH}_3\text{CH}_3$; the path in red is $\text{CH}_2 + \text{CH}_3 + \text{H} \rightarrow \text{CH}_2\text{CH}_3 + \text{H} \rightarrow \text{CH}_3\text{CH}_3$; and the path in blue is $\text{CH}_3 + \text{CH}_3 \rightarrow \text{CH}_3\text{CH}_3$.

theoretical calculations are consistent with the experimental finding that Co_2C nanoprisms formed from the initial catalyst preferentially expose facets of $\{101\}$ or $\{020\}$, which exhibit high selectivity to olefins and low selectivity to methane.

In conclusion, we have demonstrated that Co_2C nanoprisms with exposed $\{101\}$ and $\{020\}$ facets exhibit high selectivity towards the formation of lower olefins and low selectivity towards methane production under mild reaction conditions. The olefin/paraffin ratio for C_{2-4} slate is as high as 30, and the product distribution deviates markedly from the classical ASF distribution. As the facet geometry of Co_2C clearly has a very strong effect on syngas conversion, we anticipate that morphological control of the Co_2C nanostructure, with preferential exposure of $\{101\}$ and $\{020\}$, should guide the design of the next generation of highly efficient FTO catalysts.

Online Content Methods, along with any additional Extended Data display items and Source Data, are available in the online version of the paper; references unique to these sections appear only in the online paper.

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Author Contributions L.Z. and Y.S. designed the study, analysed the data and wrote the paper. F.Y. and Y.A. prepared the samples and drafted the manuscript. Y.Z. performed DFT calculations. Z.L. and T.L. studied the promoter effect. Y.L., X.Q. and Y.D. performed catalytic evaluation. L.G., J.H., S.J., Q.S. and H.W. characterized the samples. All authors discussed the results and commented on the manuscript.

Author Information Reprints and permissions information is available at www.nature.com/reprints. The authors declare no competing financial interests. Readers are welcome to comment on the online version of the paper. Correspondence and requests for materials should be addressed to L.Z. (zhongls@sari.ac.cn) or Y.S. (sunyh@sari.ac.cn).

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METHODS

Catalyst preparation. We prepared CoMn catalyst with a spinel structure ($\text{Co}_x\text{Mn}_{3-x}\text{O}_4$) by a co-precipitation method. Typically, an appropriate amount of cobalt nitrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$; Sinopharm Chemical Reagent Co.) and manganese nitrate (50 wt% $\text{Mn}(\text{NO}_3)_2$, aqueous solution; Sinopharm) were dissolved in deionized water to form a 2 M mixed salt solution (Co/Mn (mol/mol) = 2/1). Meanwhile, sodium carbonate anhydrous (Na_2CO_3 ; Sinopharm) was dissolved in deionized water, resulting in a 2 M alkali solution as precipitant. These salt and alkali solutions were simultaneously added, dropwise, into a beaker containing 100 ml deionized water under continuous stirring. A constant pH of 8.0 ± 0.1 and temperature of $30 \pm 1^\circ\text{C}$ were maintained during the precipitation process. After ageing for 2 h at 30°C , the obtained suspension was centrifuged and washed with deionized water seven times, then dried at 100°C for 10 h. The samples were then calcined in a muffle furnace at 330°C for 3 h under static air. In order to investigate the effects of sodium and manganese, we also prepared four other catalysts (Co_3O_4 , MnO_2 , CoMn-A and CoMn-Na) using similar procedures. Specifically, we prepared Co_3O_4 and MnO_2 by the same method (using Na_2CO_3 for precipitation). CoMn-A represented the CoMn catalyst ($\text{Co/Mn} = 2/1$) prepared by precipitation using $(\text{NH}_4)_2\text{CO}_3$; CoMn-Na represented the CoMn catalyst ($\text{Co/Mn} = 2/1$) prepared by co-precipitation using $(\text{NH}_4)_2\text{CO}_3$ followed by impregnation with 0.4 wt% of Na by Na_2CO_3 .

Catalyst characterization. Catalyst samples for structural analysis were removed from the reactor at different reaction times, after passivation. Specifically, the feeding gas was switched from syngas (or 10% H_2/Ar) to He and heating was stopped at selected reaction times. After the reactor cooled to room temperature (about 20°C), a flow of 1% (v/v) O_2/N_2 was introduced to passivate the catalyst for 1 h, and then the catalyst was removed from the reactor and kept in a sealed glass bottle for structural characterization.

X-ray powder diffraction patterns were used for the purpose of phase characterization and Rietveld refinement. The measurements were performed on a Rigaku Ultima IV X-ray powder diffractometer using $\text{Cu K}\alpha$ radiation with a wavelength of $1.54056 \text{ \AA}$ at 40 kV and 40 mA. Phase characterizations were realized from powder patterns using the PDF4-2015 database (Co_2C , PDF#03-065-1457; MnO , PDF#01-075-0625; Co , PDF#01-089-7093; CoO , PDF#01-071-1178). Indexing of the compounds was realized from the powder patterns using the DICVOL91 program²⁸. Quantitative Rietveld refinement was performed using the FULLPROF program. Structural data for the Co_2C , MnO and $\text{Co}_x\text{Mn}_{1-x}\text{O}$ phases were taken from the Inorganic Crystal Structure Database (ICSD, accession numbers 16895, 643195 and 9865; <https://icsd.fiz-karlsruhe.de/search/index.xhtml>). Lattice parameters for Co_2C and MnO were predetermined in a separate refinement for the 150-hour sample and kept fixed during the refinements. For all specimens, the scale factors, profile shape and broadening parameters, asymmetry and corrections for preferred orientation were refined during the quantitative analysis.

ICP analysis of sodium, cobalt and manganese was performed using an ICP optical emission spectrometer (Perkin Elmer) after sample dissolution according to standard in-house procedures.

TEM measurements were performed on a JEOL JEM 2011 electron microscope with 200 kV accelerating voltage. Samples for TEM were prepared by dispersing the powder in ethanol, followed by ultrasonication. One droplet of the suspension was dripped onto carbon-coated copper grids for measurement. We carried out EDX spectroscopy to locate the elemental distribution of cobalt, manganese and oxygen, using a silicon-drift detector on an ARM-200CF transmission electron microscope (JEOL), operated at 200 keV and equipped with double spherical aberration ('Cs') correctors. The attainable energy resolution of the EDX detector is 130 eV. Spatial drift was corrected with a simultaneous image collector.

Catalytic evaluation. Syngas conversion was carried on at 250°C in a fixed-bed reactor. Generally, 1.5 g of catalyst, in the form of a mesh of 40–60 units, mixed with 2 ml silica sand of the same size was loaded into a stainless-steel reactor with an inner diameter of 9 mm. Prior to the CO-hydrogenation reaction, the catalyst was reduced *in situ* in 10% H_2/Ar (v/v) at atmospheric pressure, with a gas-flow rate of 200 ml min^{-1} , at 300°C for 5 h; the heating ramp was 1°C min^{-1} . The temperature was then dropped to 250°C in He (99.999%) flow for 30 min, to purge the residual reduction gas. Subsequently, a mixture of 97 vol% syngas with different H_2/CO (v/v) ratios ($\text{H}_2/\text{CO} = 2, 1$, or 0.5) and 3 vol% N_2 (as an internal standard) was introduced to the reactor as feed gas.

For low-pressure testing of the CO-hydrogenation reaction, a system pressure of 1 bar was adopted. The outlet gas, after passing through the hot trap (120°C), was immediately analysed online by gas chromatography. In order to obtain detailed information on heavy hydrocarbons and oxygenates, the tail gas was allowed to pass through three absorption bottles in series, using water and toluene as absorbents, for about 24 h, and was then analysed by gas chromatography. The calculated

selectivity for oxygenates (mainly aldehydes and alcohols) was about 0.80 C% (free CO_2) when syngas with a H_2/CO ratio of 2 was used, and increased slightly to about 2 C% when the H_2/CO ratio was decreased to 1 or 0.5.

For medium- and high-pressure testing of CO hydrogenation, a system pressure of 3 bars, 5 bars or 10 bars was adopted. The outlet gas passed through a hot trap (120°C) and a cold trap (-1°C) to collect aqueous products, liquid oil products and solid wax products. Catalytic activity and product selectivities at steady state were determined by gas-chromatographic analysis of products up to C_{45} (oil products and wax products). Analysis of the products confirmed that they include CO_2 , olefins, paraffins and oxygenates. A full analysis method was used to determine the selectivity of formation of various products.

We analysed the H_2 , N_2 , CO , CH_4 , and CO_2 content of the outlet gases by using a carbon molecular sieves column (TDX-1) with a thermal conductivity detector (TCD), using helium as the carrier gas. Hydrocarbons with chains of one to ten (C_1 to C_{10}) carbons were analysed using a KCl-modified alumina capillary column (19095P-K25) with an argon carrier and a hydrogen flame ionization detector (FID). CO conversion was calculated according to an internal standard method, assuming that the amount of N_2 remained constant after the reaction. Liquid products were analysed offline with another three gas chromatographs. Aqueous products were analysed using two Porapak Q columns, furnished with an FID (for recognition of C_1 – C_5 oxygenates) or TCD (for recognition of H_2O and MeOH). Liquid oil products were analysed using an HP-1 column with N_2 carrier by FID. The mass balance, carbon balance and oxygen balance calculated in each test was between 97% and 101%.

CO conversion was calculated on a carbon-atom basis, as follows:

$$\text{CO conversion} = \frac{\text{CO}_{\text{inlet}} - \text{CO}_{\text{outlet}}}{\text{CO}_{\text{inlet}}} \times 100\%$$

where CO_{inlet} and $\text{CO}_{\text{outlet}}$ are moles of CO at the inlet and outlet, respectively.

CO_2 selectivity was calculated according to:

$$\text{CO}_2 \text{ selectivity} = \frac{\text{CO}_2 \text{ outlet}}{\text{CO}_{\text{inlet}} - \text{CO}_{\text{outlet}}} \times 100\%$$

where $\text{CO}_2 \text{ outlet}$ refers to moles of CO_2 at the outlet.

The selectivity of an individual product (hydrocarbon or oxygenate), C_nH_m , in a CO_2 -free reaction was obtained according to:

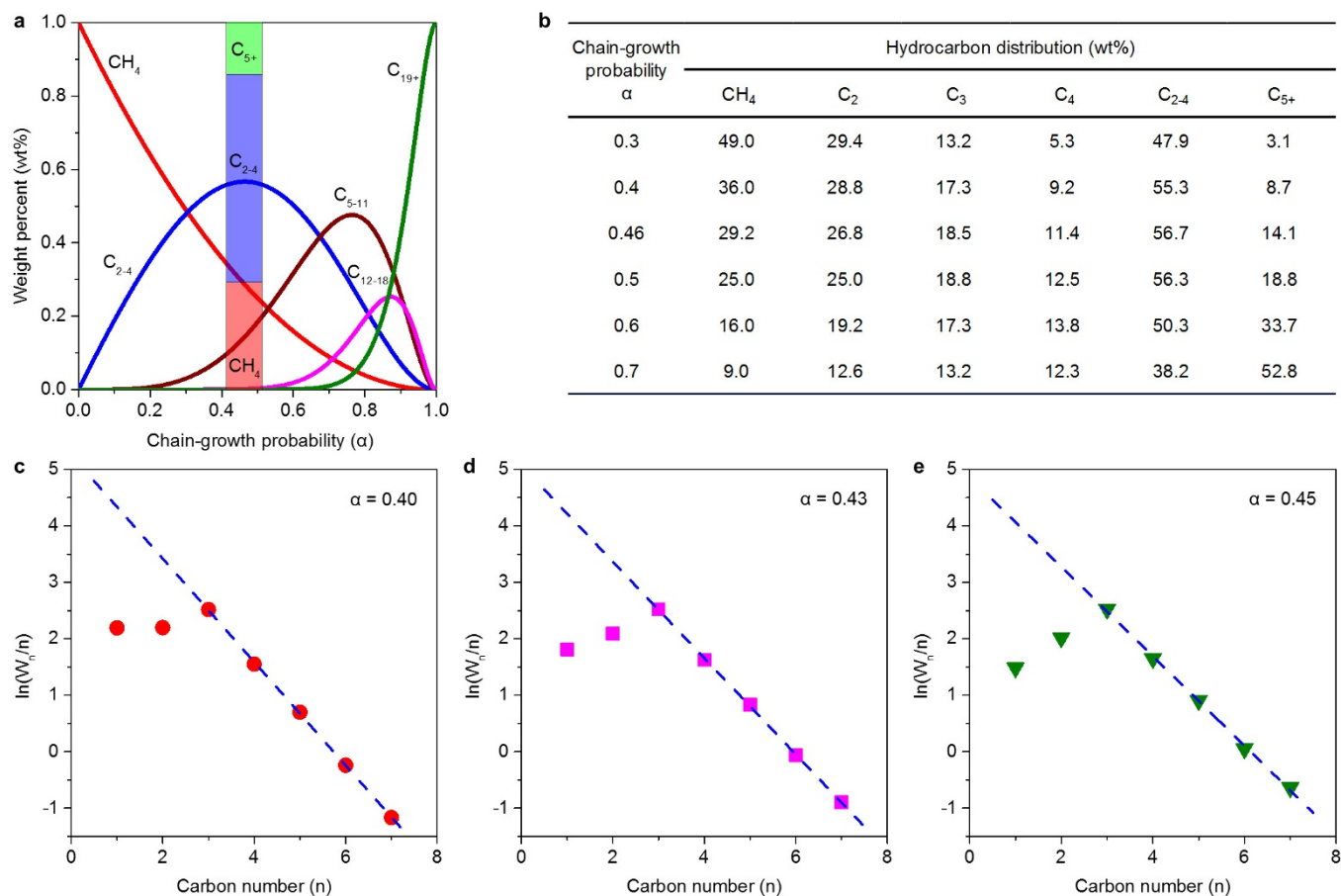
$$\text{C}_n\text{H}_m \text{ selectivity} = \frac{n\text{C}_n\text{H}_m \text{ outlet}}{\text{CO}_{\text{inlet}} - \text{CO}_{\text{outlet}} - \text{CO}_2 \text{ outlet}} \times 100\%$$

Where $\text{C}_n\text{H}_m \text{ outlet}$ represents moles of individual product (hydrocarbon or oxygenate) at the outlet.

The catalytic performance changed during the initial stage. It was hard to fully analyse all products in the short time period (1 h) at 1 bar, as it would take a longer time for tail-gas treatment with the absorption bottles. In this case, we used a difference method to calculate the selectivity for the production of C_{5+} hydrocarbons (those with five or more carbons) plus oxygenates.

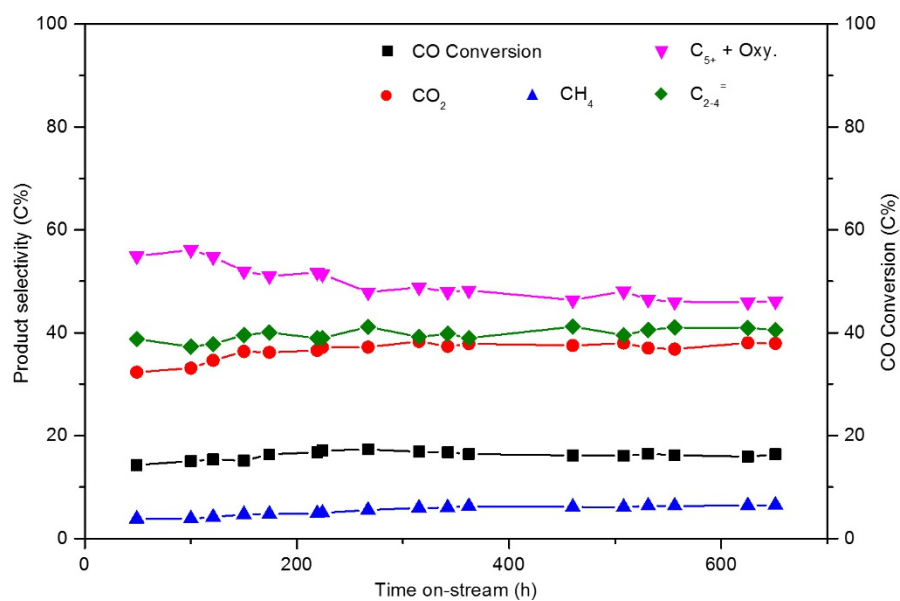
Four repeat experiments were carried out under the same reaction conditions. We found that the catalytic performance of the studied catalyst showed good reproducibility, with both CO conversion and CO_2 selectivity fluctuating within 5%, and the fluctuation in each hydrocarbon selectivity (CO_2 -free) being less than 2%. **Computations.** We performed spin-polarized DFT calculations based on DFT as implemented in the Vienna *ab initio* simulation package (VASP)^{29,30} using the projector augmented wave (PAW)^{31,32} method. We used Perdew–Burke–Ernzerhof (PBE) functionals³³ to evaluate the non-local exchange–correlation energy, with a planewave energy cutoff of 400 eV. The vacuum region between periodically repeated slabs was 15 Å, and force convergence was 0.02 eV. $p(2 \times 1)$, $p(2 \times 2)$, $p(2 \times 2)$ and $p(3 \times 3)$ unit cells—for $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$, $\text{Co}_2\text{C}(111)$ and $\text{Co}(0001)$ surfaces, respectively—were used in our calculations. A $p(3 \times 3)$ unit cell was applied for the $\text{Co}(0001)$ surface. The topmost two equivalent Co layers, including C atoms, are fully relaxed for all of the Co_2C surfaces, while the top three layers of the $\text{Co}(0001)$ surface are allowed to relax in all calculations. The adsorbates are also allowed to be fully relaxed throughout. Monkhorst–Pack³⁴ k -point samplings of $(3 \times 4 \times 1)$, $(3 \times 2 \times 1)$, $(3 \times 4 \times 1)$ and $(3 \times 3 \times 1)$ were used in calculations of the $\text{Co}_2\text{C}(101)$, $\text{Co}_2\text{C}(020)$, $\text{Co}_2\text{C}(111)$ and $\text{Co}(0001)$ surfaces, respectively. Transition states were located using the force reversed method³⁵, and a force tolerance of $0.02 \text{ eV \AA}^{-1}$ was used without zero-point energy correction. Transition states of some of the minimum-energy reaction pathways were located by using the climbing image nudged elastic band (CI-NEB) method^{36,37}. Separate adsorption of the species involved in the reactions was adopted for calculation of activation barrier and reaction energy.

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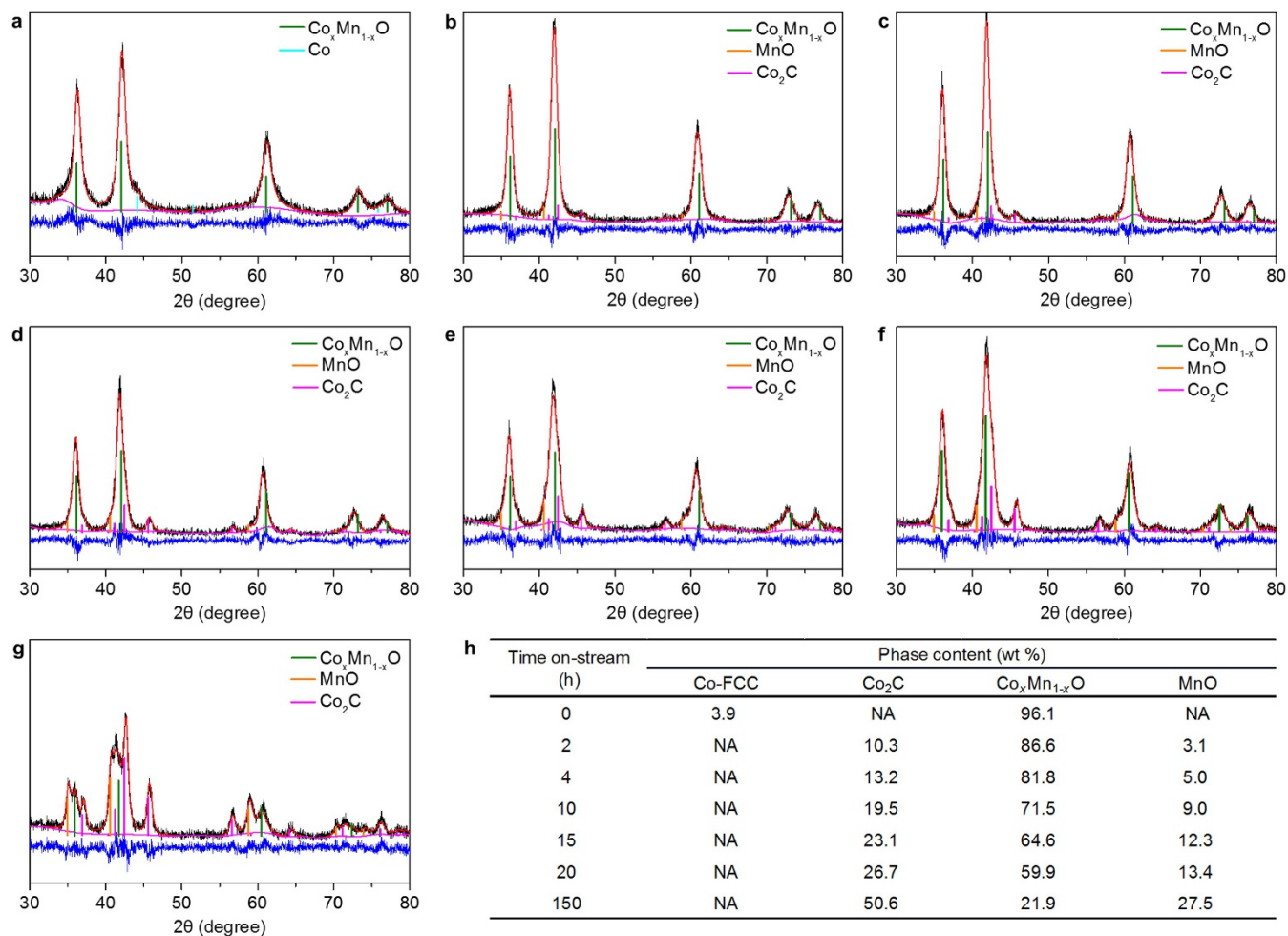


Extended Data Figure 1 | Distribution of products generated by the FTO process from syngas. **a**, Product distribution according to the Anderson–Schulz–Flory (ASF) model. The vertical bar shows the calculated hydrocarbon distribution for a chain-growth probability (α) of 0.46, which provides the highest proportion of C₂–C₄ hydrocarbons. **b**, Hydrocarbon distribution according to the ASF model for chain-growth

probabilities of 0.3, 0.4, 0.46, 0.5, 0.6 and 0.7. **c–e**, Typical product plots obtained using our CoMn catalyst (showing $\ln(W_n/n)$ versus n) when the reaction was performed under a H₂/CO ratio of 2 (**c**), 1 (**d**) or 0.5 (**e**), at a temperature of 250 °C and a pressure of 1 bar. The chain-growth probabilities were obtained by fitting the results obtained for C_{3–7} using the ASF model.



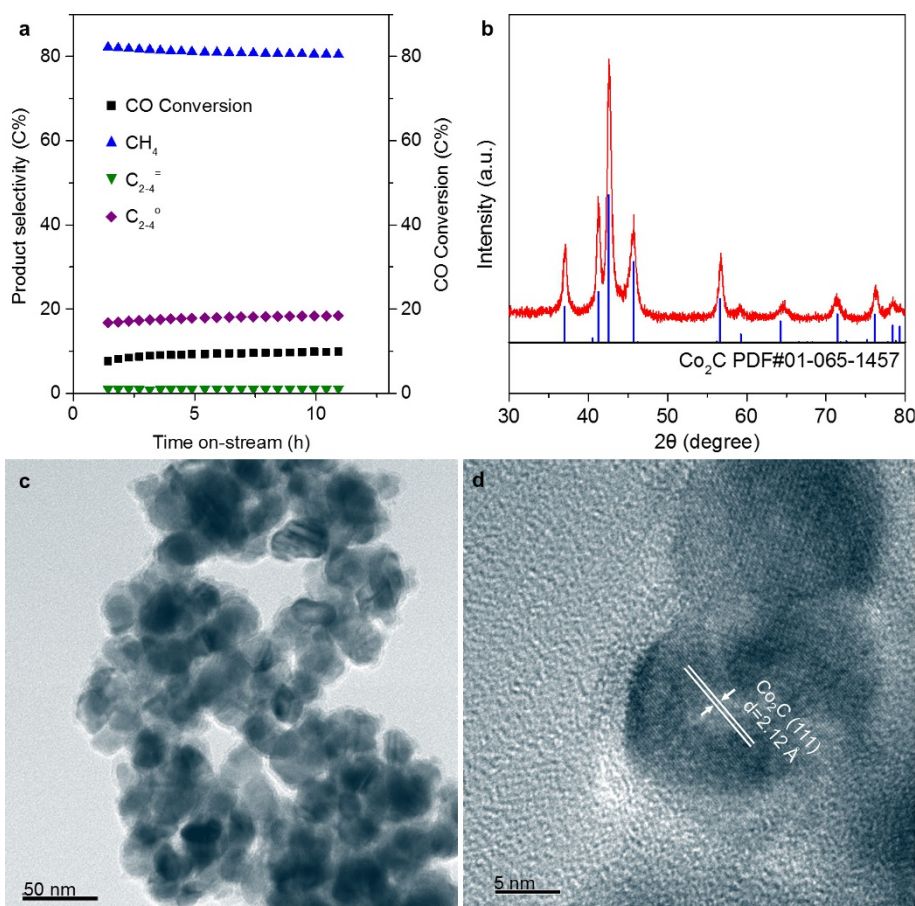
Extended Data Figure 2 | Stability test for the CoMn catalyst. Reaction conditions: 250 °C, 3 bar, 6,000 ml h⁻¹ g_{cat}⁻¹, H₂/CO = 1. The selectivity of the indicated products remains more or less constant over more than 600 h.



Extended Data Figure 3 | X-ray diffraction analysis of the CoMn catalyst at different times on-stream. **a–g**, Results from refinement of the X-ray diffraction patterns of catalysts at 0 h (**a**), 2 h (**b**), 4 h (**c**), 10 h (**d**), 15 h (**e**), 20 h (**f**) and 150 h (**g**) ('0 h' refers to the catalyst after

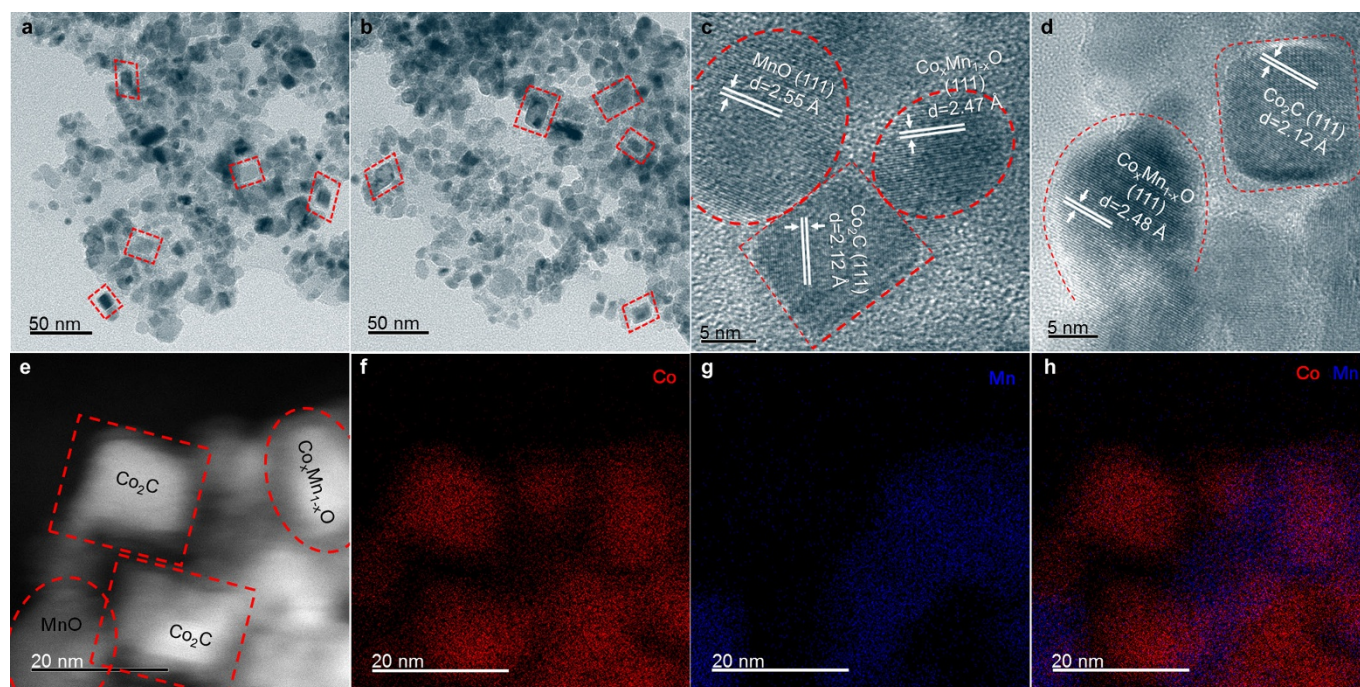
reduction). The graphs show the different phases of the catalyst (Co , Co_2C , $\text{Co}_x\text{Mn}_{1-x}\text{O}$ and MnO) that are present at different times on-stream.

h, Quantification of the refinement results on the basis of a full Rietveld analysis. NA, not available.



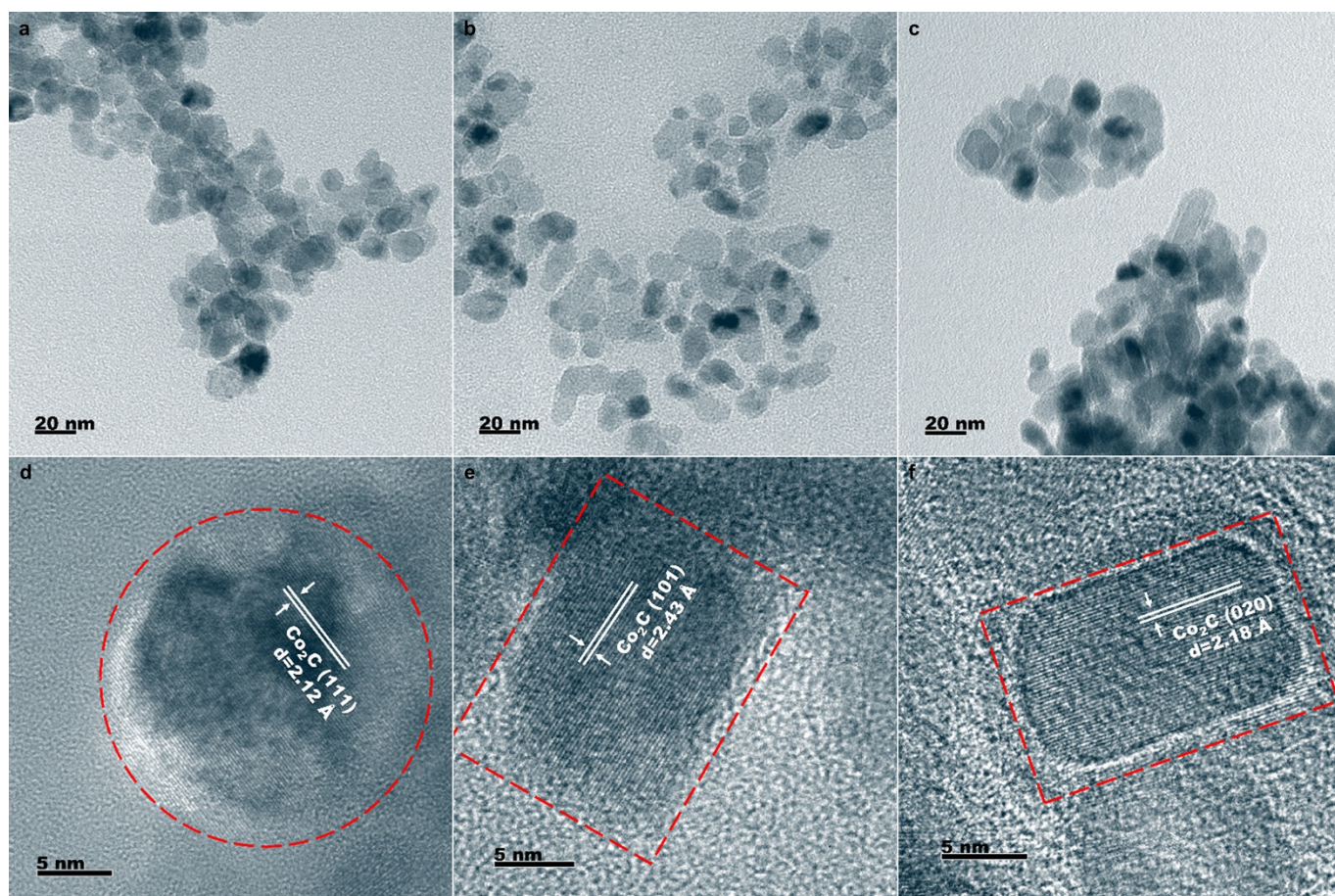
Extended Data Figure 4 | Catalytic performance and structure of Co_2C sphere-like nanoparticles. **a**, Catalytic performance of Co_2C sphere-like nanoparticles with time on-stream. **b**, XRD pattern, **c**, TEM image and **d**, high-resolution TEM image of Co_2C sphere-like nanoparticles. The Co_2C was prepared by carbonizing Co_3O_4 with pure CO at a temperature of 250 °C and at atmospheric pressure for 24 h. The reaction was performed at 250 °C, 1 bar, 2,000 $\text{ml h}^{-1} \text{ g}_{\text{cat}}^{-1}$, $\text{H}_2/\text{CO} = 2$. The calculated

reaction rate for such Co_2C sphere-like nanoparticles was $2.8 \times 10^{-3} \text{ mol CO h}^{-1} \text{ g}_{\text{cat}}^{-1}$ for a CO conversion of 9.5%. The calculated reaction rate for the Co_2C catalyst in ref. 18 was $8.9 \times 10^{-4} \text{ mol CO h}^{-1} \text{ g}_{\text{cat}}^{-1}$ (CO conversion of 2%, at 20 bar, 220 °C, 3,000 $\text{ml h}^{-1} \text{ g}_{\text{cat}}^{-1}$, $\text{H}_2/\text{CO} = 2$). The calculated reaction rate for the studied Co_2C nanoprism catalyst was $9.4 \times 10^{-3} \text{ mol CO h}^{-1} \text{ g}_{\text{cat}}^{-1}$ (CO conversion of 31.8% at 250 °C, 1 bar, 2,000 $\text{ml h}^{-1} \text{ g}_{\text{cat}}^{-1}$, $\text{H}_2/\text{CO} = 2$).



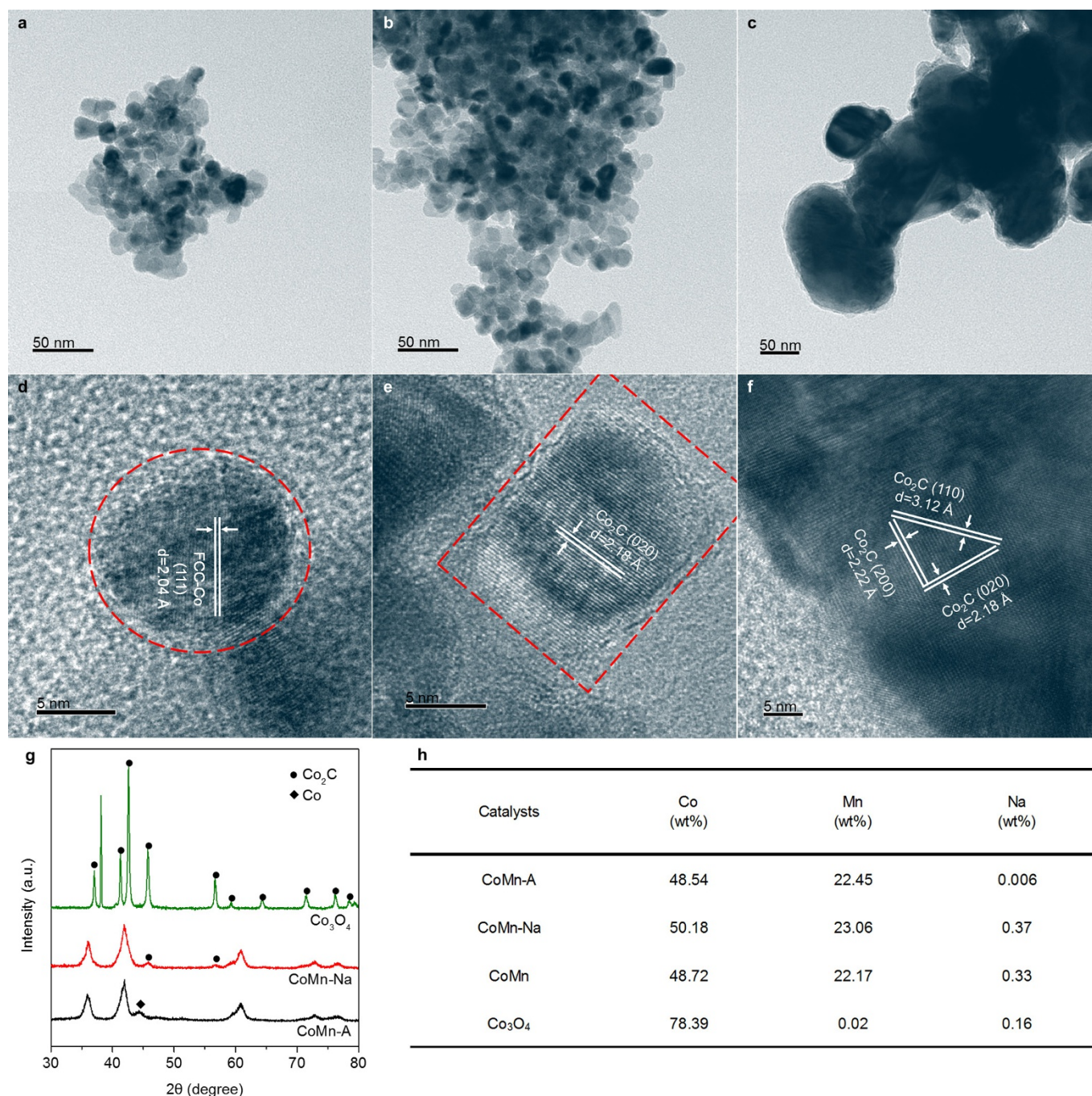
Extended Data Figure 5 | TEM images of the CoMn catalyst after reaching steady state. **a–d**, Low-resolution images (**a**, **b**) and high-resolution images (**c**, **d**) showing Co_2C nanoprisms (parallelepiped structures) and sphere-like nanoparticles of MnO or $\text{Co}_x\text{Mn}_{1-x}\text{O}$. **e**, Scanning TEM image. **f–h**, EDX mapping of Co (**f**), Mn (**g**) and Co plus Mn (**h**). For the nanoprism particles, a higher concentration of Co and a very low concentration of Mn was observed; these results, coupled

with the high-resolution TEM images of the lattice fringes, suggest that such nanoprisms are composed of Co_2C . For most of the sphere-like nanoparticles, both Co and Mn were observed, indicating that these particles are composed of CoMn composite oxide ($\text{Co}_x\text{Mn}_{1-x}\text{O}_y$). For some of the sphere-like nanoparticles, only Mn was observed and the concentration of Co was very low, suggesting that these particles are composed of MnO.



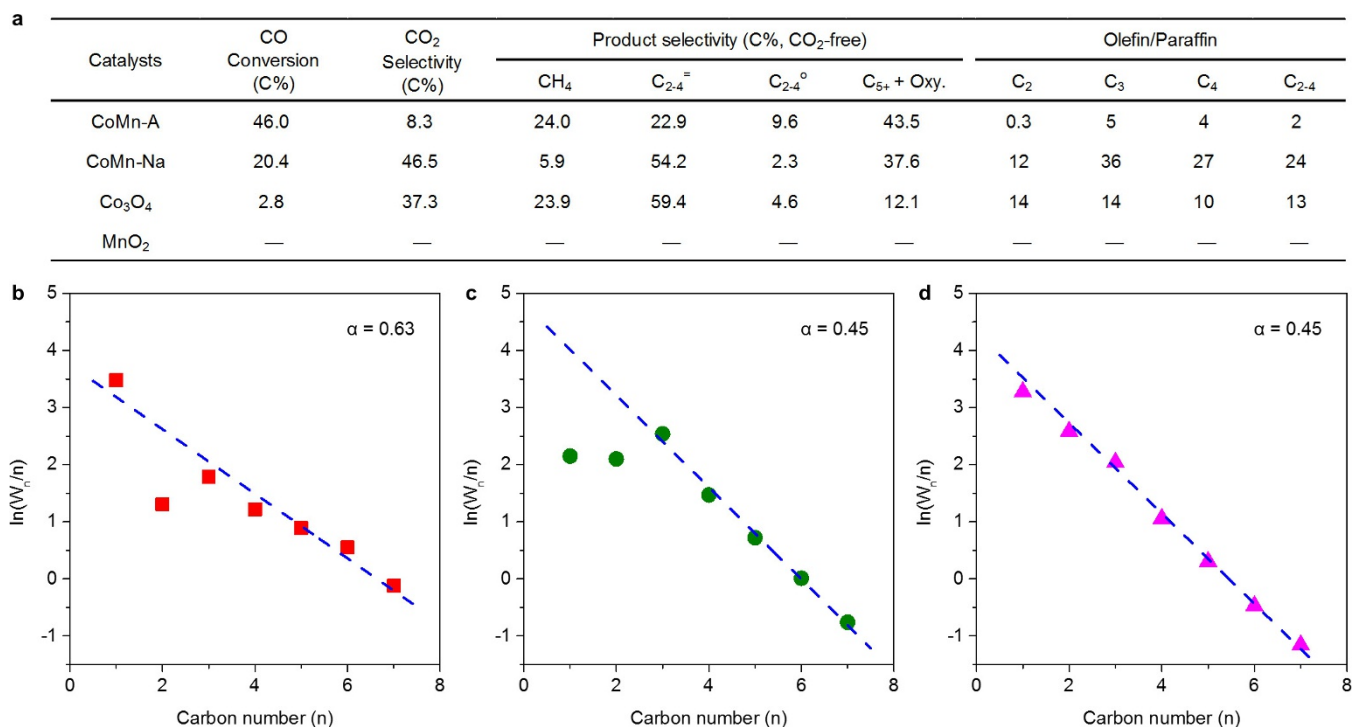
Extended Data Figure 6 | TEM images of the CoMn catalyst at different reaction times. **a–c**, Low-resolution TEM images at 2 h (**a**), 10 h (**b**) and 20 h (**c**). **d–f**, Corresponding high-resolution TEM images. For the samples removed at 2 h, it was hard to find Co₂C nanoprisms; the shape of the

Co₂C seemed to be irregular. At 10 h, the Co₂C nanostructure seemed to be nanoprism-like, with a parallelepiped shape, and the lattice fringes suggested the exposed facet of (101) geometry, although the shape was not perfect. At 20 h, Co₂C nanoprisms were found.



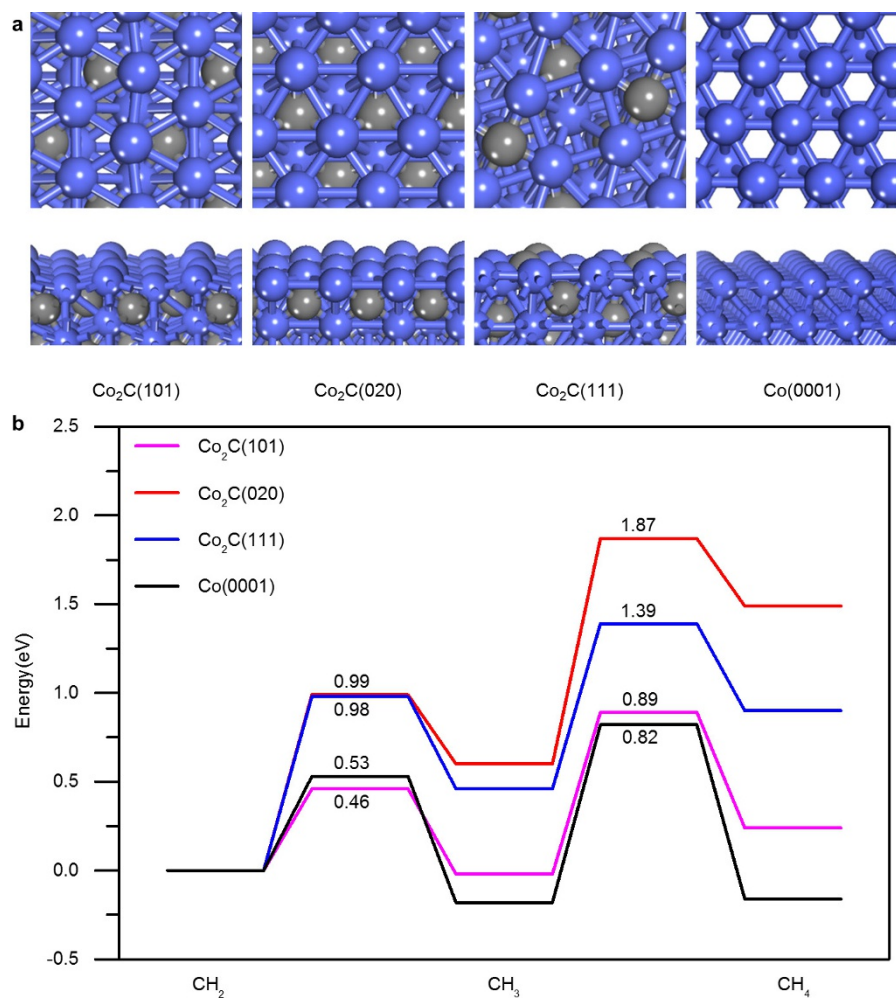
Extended Data Figure 7 | TEM, XRD and ICP analyses of the effects of Na and Mn on different Co-based catalysts. a–c, Low-resolution TEM images of spent CoMn-A (a), CoMn-Na (b) and Co₃O₄ (c). **d–f,** Corresponding high-resolution TEM images. **g,** XRD patterns of spent CoMn-A, CoMn-Na and Co₃O₄ catalysts. **h,** Elemental analysis of the fresh catalysts by ICP mass spectrometry. To produce CoMn-A, CoMn catalyst (Co/Mn = 2/1) was precipitated with (NH₄)₂CO₃. To

produce CoMn-Na, CoMn catalyst (Co/Mn = 2/1) was precipitated with (NH₄)₂CO₃ and impregnated with about 0.4 wt% of Na using Na₂CO₃. CoMn was prepared by precipitating CoMn catalyst (Co/Mn = 2/1) with Na₂CO₃. Co₃O₄ was prepared by precipitation using Na₂CO₃. We found CoMn-A to include sphere-like, face-centred-cubic, metallic Co nanoparticles. We detected Co₂C nanoprisms in CoMn-Na, while Co₃O₄ comprised larger Co₂C sphere-like nanoparticles.



Extended Data Figure 8 | Catalytic performance and product distribution of different catalysts. **a**, Performance of different catalysts. **b–d**, Product plot ($\ln(W_n/n)$ versus n) for CoMn-A (**b**), CoMn-Na (**c**) and Co₃O₄ (**d**) catalysts. CoMn-A, CoMn-Na and Co₃O₄ were prepared as

described in Extended Data Fig. 7. MnO₂ was prepared by precipitation using Na₂CO₃. Reaction conditions: 1 bar, 250 °C, 2,000 ml h^{−1} g_{cat}^{−1}, H₂/CO = 2. There was no detectable CO conversion by MnO₂.



Extended Data Figure 9 | DFT study on different surfaces. **a**, Top views (upper panels) and side views (bottom panels) of different surfaces: from left to right, Co₂C(101), Co₂C(020), Co₂C(111) and Co(0001). Blue, Co atom; grey, C atom. **b**, Energy profiles for CH₄ formation on Co₂C(101), Co₂C(020), Co₂C(111) and Co(0001) surfaces. The intermediate state of CH₂ + 2H is chosen as the zero point for all of the energy profiles.

Extended Data Table 1 | Catalytic performance of CoMn catalyst in the initial stages of the FTO reaction

Time on stream (h)	CO Conversion (C%)	CO ₂ Selectivity (C%)	Product selectivity (C%, CO ₂ -free)				Olefin/Paraffin			
			CH ₄	C ₂₋₄ ⁼	C ₂₋₄ ^o	C ₅₊ + Oxy.	C ₂	C ₃	C ₄	C ₂₋₄
0.2	25.3	13.9	1.8	11.8	1.0	85.4	5	24	15	12
1.9	10.3	27.2	4.9	14.8	1.1	79.2	7	25	17	14
4.2	6.8	35.8	4.7	19.7	0.9	74.7	16	32	20	22
10.4	15.5	47.1	5.6	46.7	1.5	46.3	23	44	28	32
15.3	21.7	45.6	5.3	53.3	1.6	39.8	23	45	30	34
20.2	23.6	45.4	5.2	55.2	1.6	37.9	22	45	30	34

Reaction conditions: 250 °C, 1 bar, 2,000 ml h⁻¹ g_{cat}⁻¹, H₂/CO = 2.